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MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: B. Tech. Biotechnology Semester: V

Name of the Paper: Stem Cell Biology

Paper Code: TBT504

Time: 90 Minutes

Maximum Marks: 50

Note:-

- (i) Answer **all the questions** by choosing **any one of the two sub questions**.
(ii) Each question carries 10 marks

Q1	(10marks)	CO1
(a)	Review the milestones of advances in stem cell biology.	
OR		
(b)	Define stem cells and discuss the general properties of stem cells.	CO1, CO2
Q2	(10 marks)	
(a)	Describe embryonic stem cells with emphasis on its production and maintenance.	
OR		CO1, CO2
(b)	Explain the different methods for fate mapping of stem cells.	
Q3	(10 marks)	
(a)	Discuss adult stem cells and its applications.	CO1, CO2
OR		
(b)	Elaborate on cancer stem cells and the different signaling pathways.	
Q4	(10 marks)	CO2
(a)	Review mesenchymal stem cells and its differentiation potential.	
OR		
(b)	Explain induced pluripotent stem cells.	CO2
Q5	(10 marks)	
(a)	Describe neural stem cells and its role in neurogenesis.	
OR		CO2
(b)	Briefly describe the tests used for the identification of stem cells.	

